

8 SEPT 2025

Quick Work

Page No.:

Date:

M T W T F S S

Multiplexing:

- Technique to combine multiple signals in one medium

- Purpose: Efficient use of bandwidth

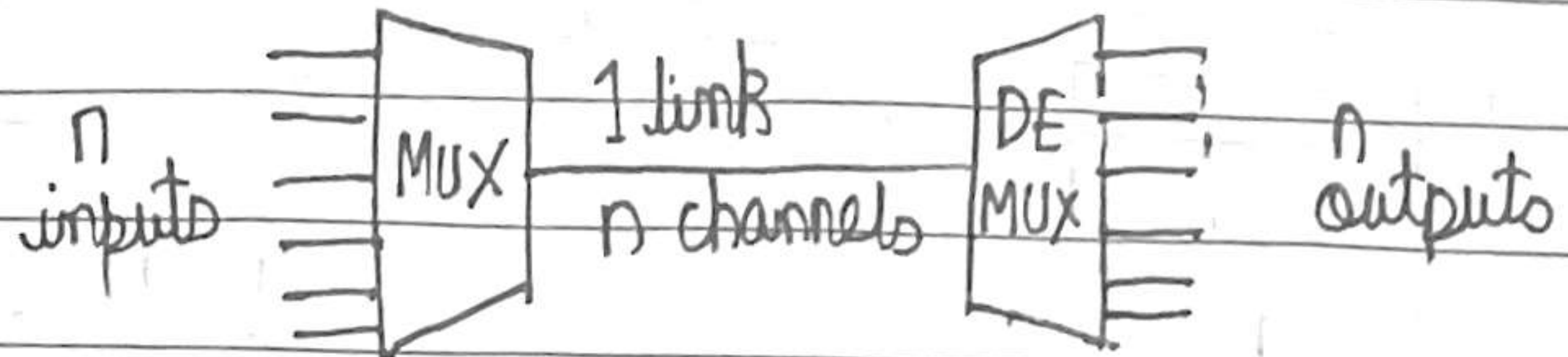
- Type:

Frequency

Time

Wavelength

Spread Spectrum



- Applications:

(i) Telecommunications

(ii) Television Broadcasting

(iii) Internet and Data communication

(iv) Optical Fiber Communication

(v) Satellite Communication

(vi) Wireless Communication

(vii) Sensor Networks and IoT

- Need for MUX:

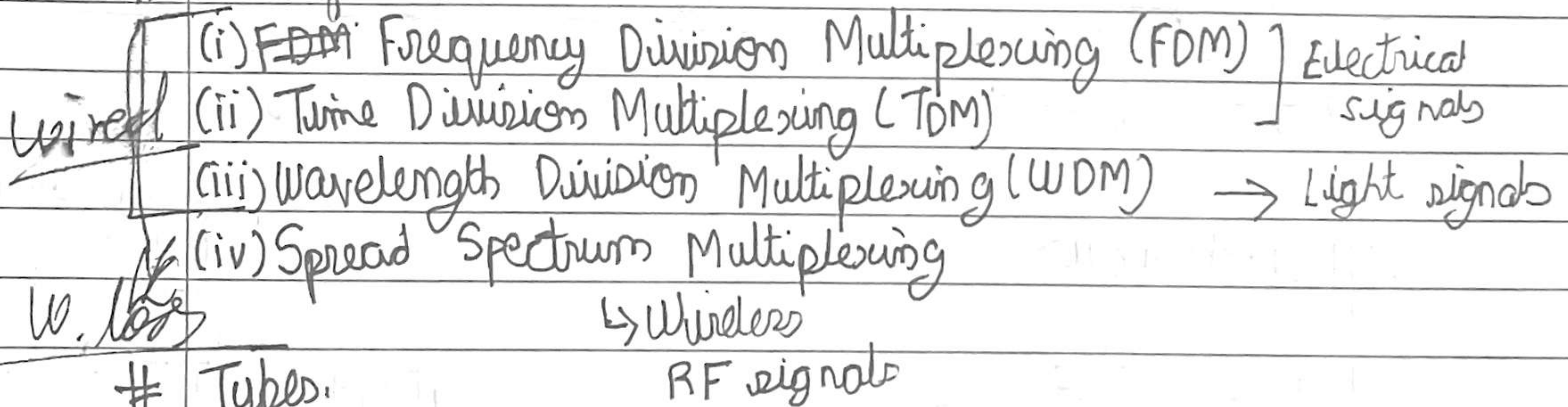
(i) Scarcity of transmission resources

(ii) Multiple users or services

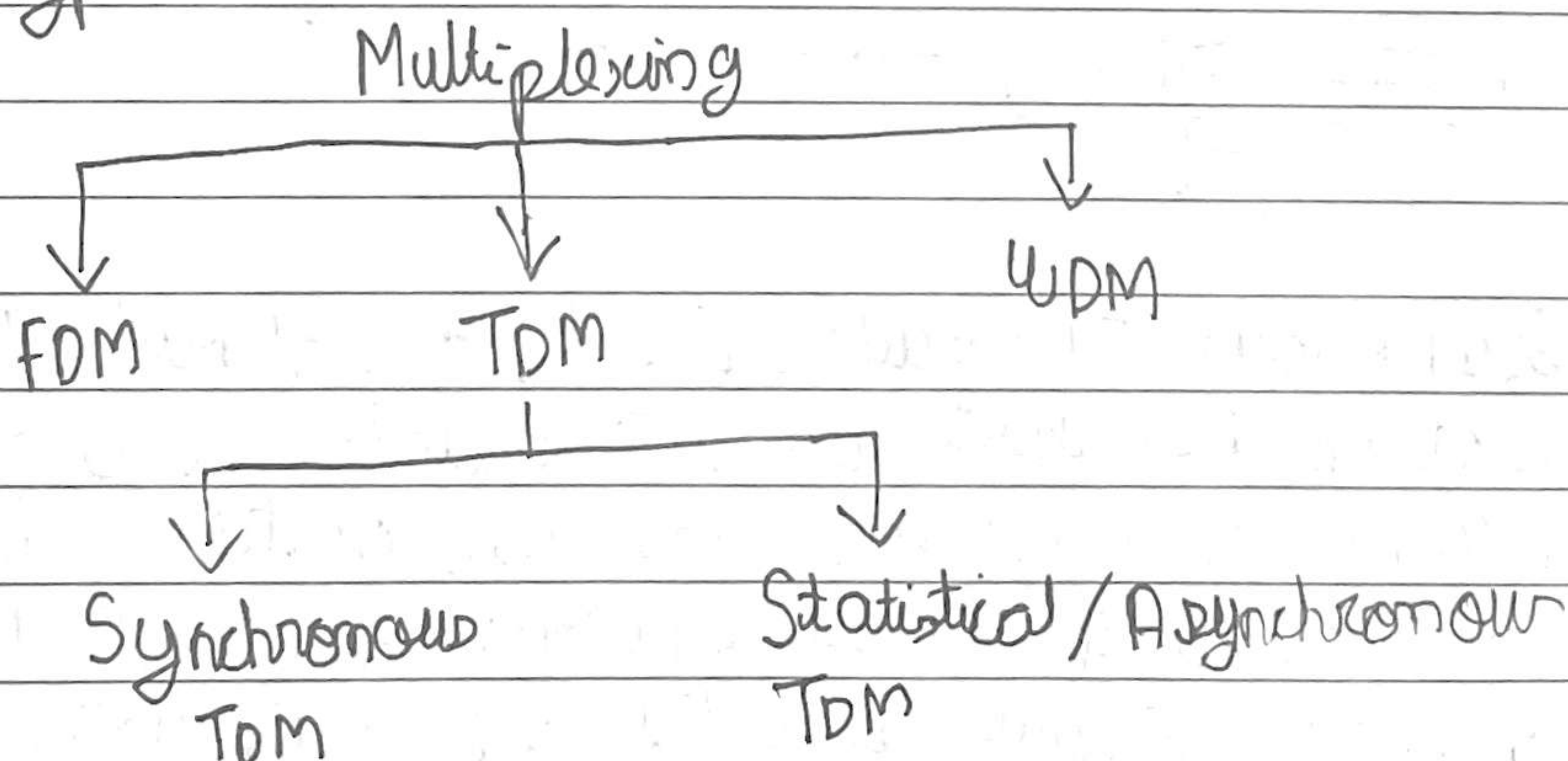
(iii) Cost efficiency

(iv) Improved channel utilization

Classification



Types:

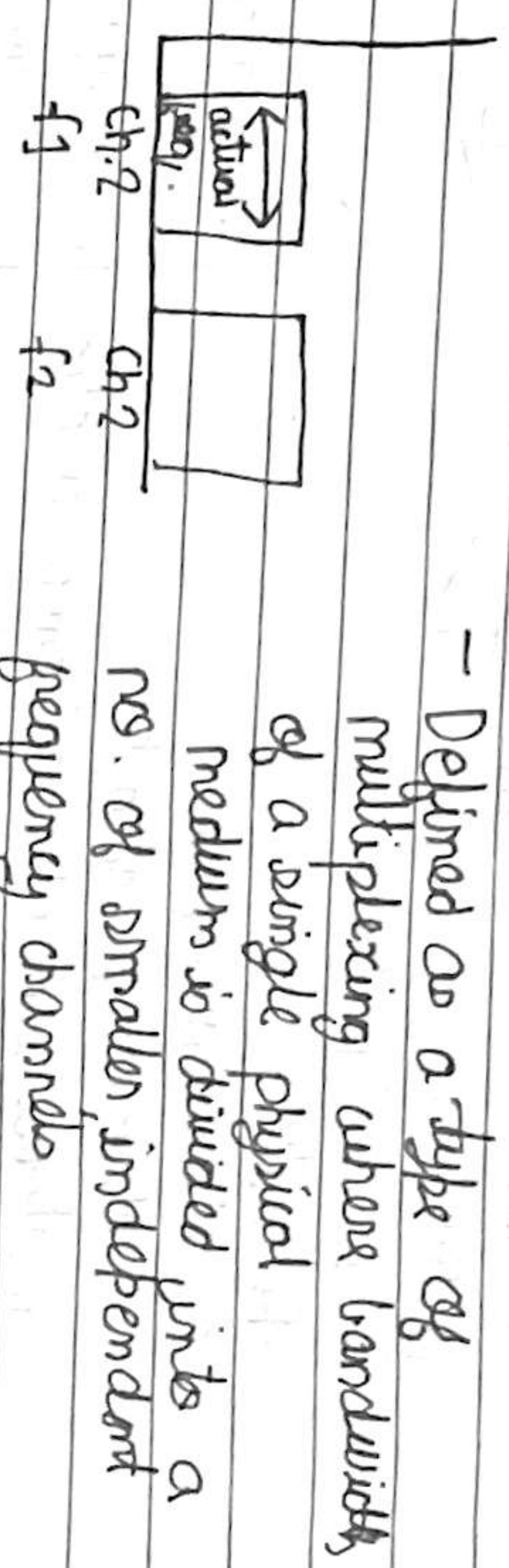
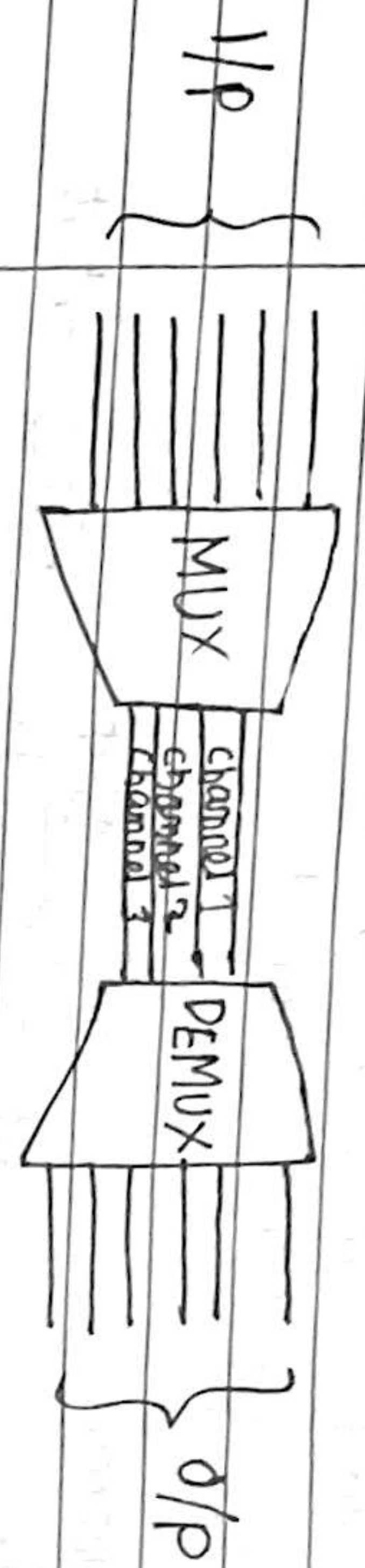


BSS = Bandwidth Spread

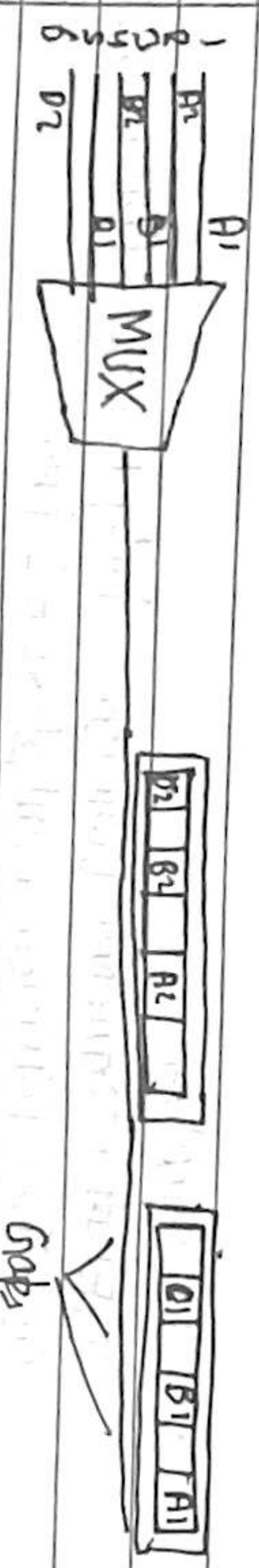
FDM

FE R = Frequency Error Rating

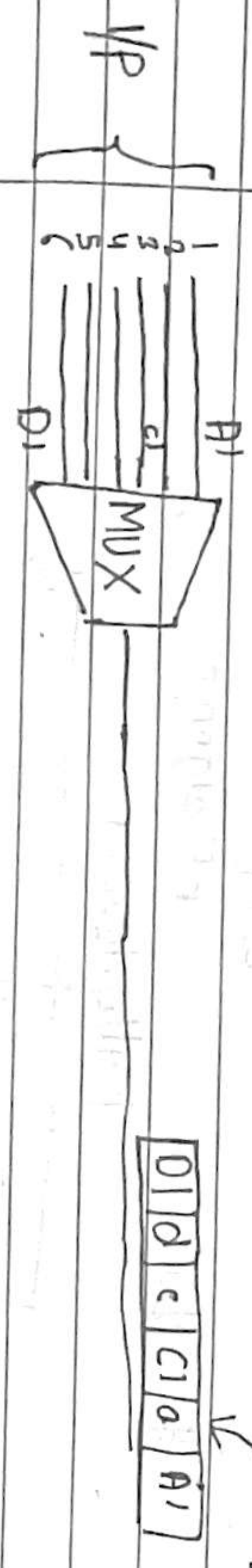
Quick Work
Page No.:
Date: M T W T F S S



TDM (Instead of sharing a portion of bandwidth in synch. remod form of channels, time is shared)



(b) Asynchronous



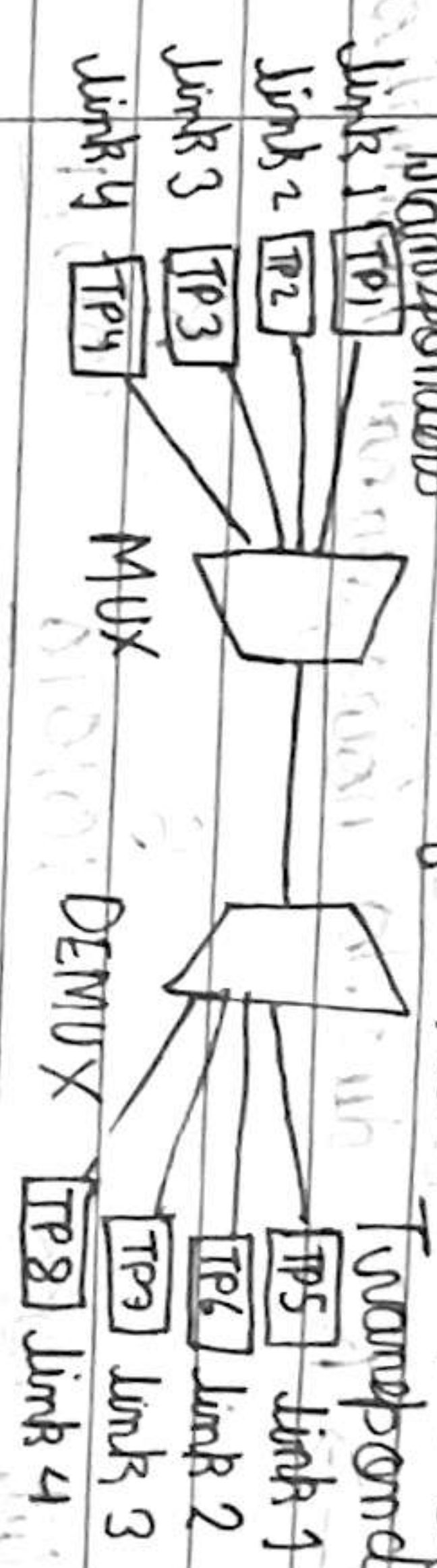
Synchronous TDM = where input frame already has a slot in the output frame. Time slots are grouped into frames. One frame = one cycle of time slots.
Asynchronous TDM = where output frame collects data from input frame till it is full, not leaving an empty slot. We include address of each particular data in slot.

12 SEP 2025

Quick Work
Page No.:
Date: M T W T F S S

#

WDM - Used to increase capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.



signal flow

Advantages of Multiplexing -

- Efficient use of bandwidth.
- ↑ data transmission.
- Scalability.
- Flexibility.

Disadvantages of Multiplexing -

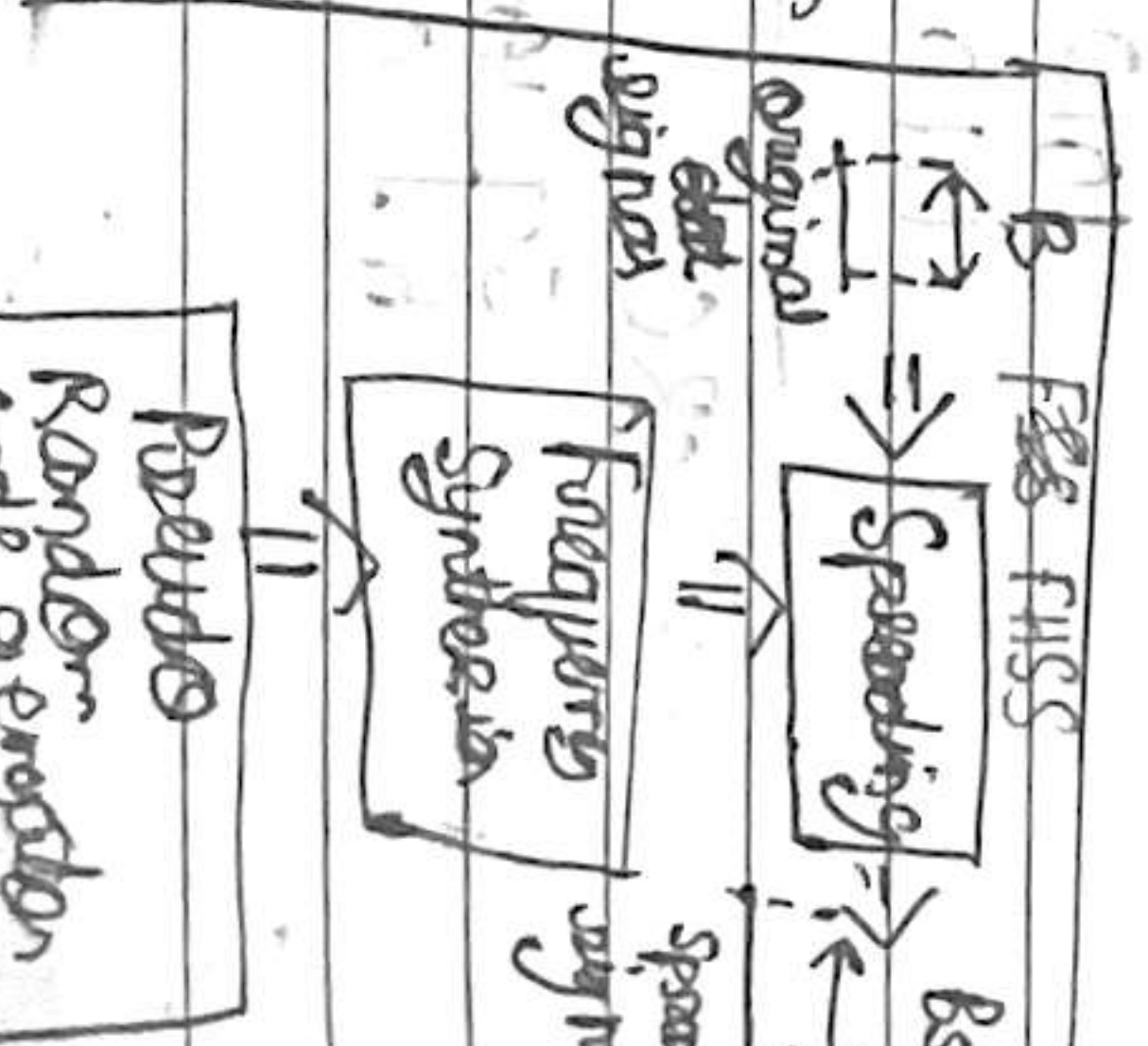
- Synchronization issues.
- Latency.
- Signal degradation.
- Resource management.

Spread Spectrum -

- Wireless communication technology that distributes the transmitted signal across a larger bandwidth than the original signal.

Types Techniques:
(i) Frequency Hopping Spread Spectrum
(ii) Direct Sequence Spread Spectrum
Differant carrier frequencies are modulated by same signal i.e. M carrier frequencies are modulated by the signal.

DSSS = Bandwidth of the original signal is also expanded by a different technique. Each bit replaced with n bits using spreading code called chips. $\text{chip rate} = n \times \text{bit rate}$. $\text{Spreading} \Rightarrow \text{Spread signal}$.



$$S + r = 2^r$$

Hamming Code / Distance

HC

$$d = \text{data} = 1100$$

$$r = \text{redundancy} = 2^r \geq d + 1 + 1$$

$$\text{Total length} = d + r$$

1100

$$r = 3$$

$$2^3 \geq 4 + 3 + 1$$

Satisfied

$$\text{Total Length} = 4 + 3 = 7 \text{ bits}$$



7 6 5 4 3 2 1

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$



D7 D6 D5 D4 R4 R3 R2

0111 0101 0011 0001

$$R_1 = 7$$

$$R_2 = 2$$

only see where last bit is

D7 D5 D3 R1

1 0 0 R1

list

add

To make it even, we are going to consider

R1 as "1"



D7 D6 D5 D4 R4 R3 R2

Now, where 2nd last bit is 1



D7 D6 D5 D4 R4 R3 R2

01 D6 D5 R4 D3 R2 R1

0111 0110 0011 0010

D7 D6 D4 D3 R2

1 1 0 R2

Already even

$$\Rightarrow R2 = 0$$

Similarly, now check where 3rd last bit is 1

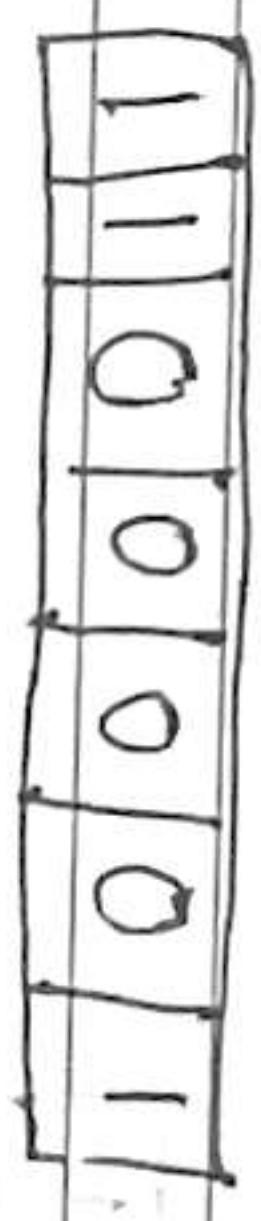


D7 D6 D5 D4 R4 D3 R2 R1

0111 0101 0100

D7 D6 D5 R4 Already even

1 1 0 R4 = 0



Now at receiver's side



We have deliberately used 1 instead of 0 at D3 to show the error

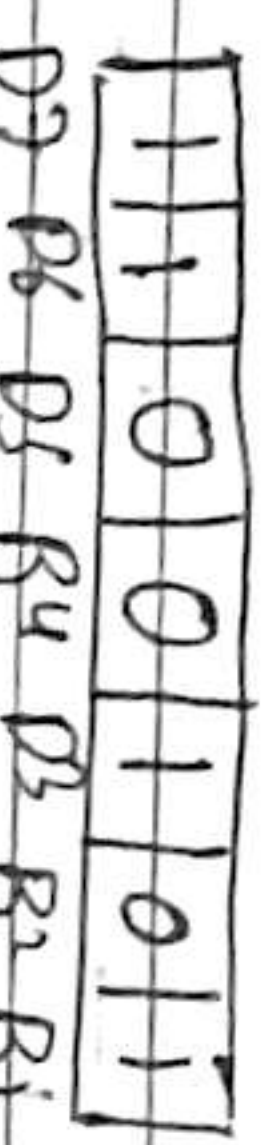
D7 D6 D5 R4 D3 R2 R1
0111 0101 0011 0001

Find R1

D7 D5 D3 R1 $\Rightarrow R1 = 1$
1 0 1 1

add as

Find R2



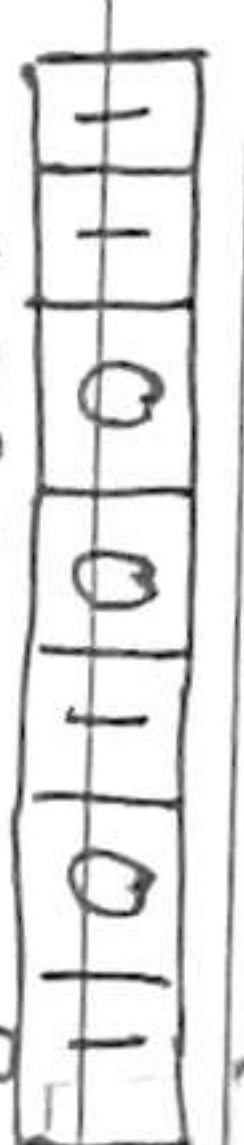
D7 D6 D5 R4 D3 R2 R1
0111 0110 0011 0010

When even parity is there for R_n ,
else, $R_n = 1$.
Here $R_n = 0$

D7 D6 D3 R

Then write it sequentially like $R_4 R_2 R_1$.
The binary no. formed will be the error position

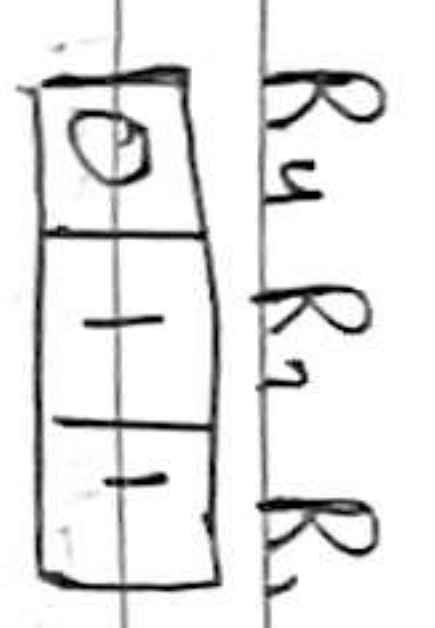
Find R4



D7 D6 D5 R4 D3 R2 R1
0111 0110 0101 0100

D7 D6 D3 R4 R4 = 0

1 0 0 0 even only



011 = 3

error is at 3rd position

Invert it to correct the error



D7 D6 D5 R4 D3 R2 R1

Now, check again for the final time

For $R_1 \Rightarrow$ 1 0 1 1

even bit $R_1 = 0$

~~D7 D6 D5~~

D7 D6 D3 R2 even
1 1 0 0

$R_1 = 0$
 $R_2 = 0$

For $R_4 \Rightarrow$

D7 D6 D5 R4 even
1 1 0 0

$R_1 = 0$
 $R_2 = 0$
 $R_4 = 0$

R4 R2 R1

0 0 0 No error!

#

Block coding -

Enable system to detect & correct both single & burst/multiple bit errors

Block coding can detect multiple bit errors, but typically can correct only one bit at a time

1 Oct 2025

Channel All

- Decides how to allocate a single broadcast channel among competing users
- The channel might be a portion of the wireless

Summary Table -

Technique	Domain	Used for	Key Feature
FDM	Analog	Radio, TV, cable	Frequency based sep.
TDM	Digital	Digital voice/data	Time based separation
WDM	Optical	Fiber optic	Wavelength separation
DSSS	Wireless/digital	CDMA, Wi-Fi	Signal spreading with spread
FHSS	Wireless	Bluetooth	Frequency hopping

Spread Spectrum -

- Transmits signals over a wide frequency range
- Spreads bandwidth beyond what's needed for the data
- Provides resistance to interference, better security and privacy
- Uses a spread code to expand the signal's bandwidth
- Commonly used for secure communication to prevent interception to prevent jamming

Principles -

- Bandwidth is used is much larger than needed for redundancy
- Spreading occurs after signal is generated

Characteristics -

- Higher channel capacity
- Resistant to jamming
- Easily intercept any unauthorized person
- Ability to resist multipath propagation

FHSS -

1) Advantages -

- Synchronization is not greatly dependent on distance
- Processing gain is higher than DSSS

2) Disadvantages -

- Bandwidth is too large (GHz)
- Complex & expensive digit frequency synthesizers are required

3) Applications -

- Used in Bluetooth
- Military communications
- WLANs
- UWB (Ultra Wide Band)
- Remote Controls

SSS DSSS -

1) Advantages -

- Combats jamming most effectively
- Performance in presence of noise is superior to FHSS
- Interference is minimized against the signal

2) Disadvantages -

- Processing gain is lower than FHSS
- Channel Bandwidth is less than FHSS
- Synchronization is affected by variable distance between the transmitter & receiver

3) Applications -

- GPS (Global Positioning System)
- CDMA (Code Division Multiple Access) Cellular Networks
- Satellite communication
- Wireless Sensor Networks

ARQ = Automatic Repeat reQuest

Flow Control and Error control Protocols.

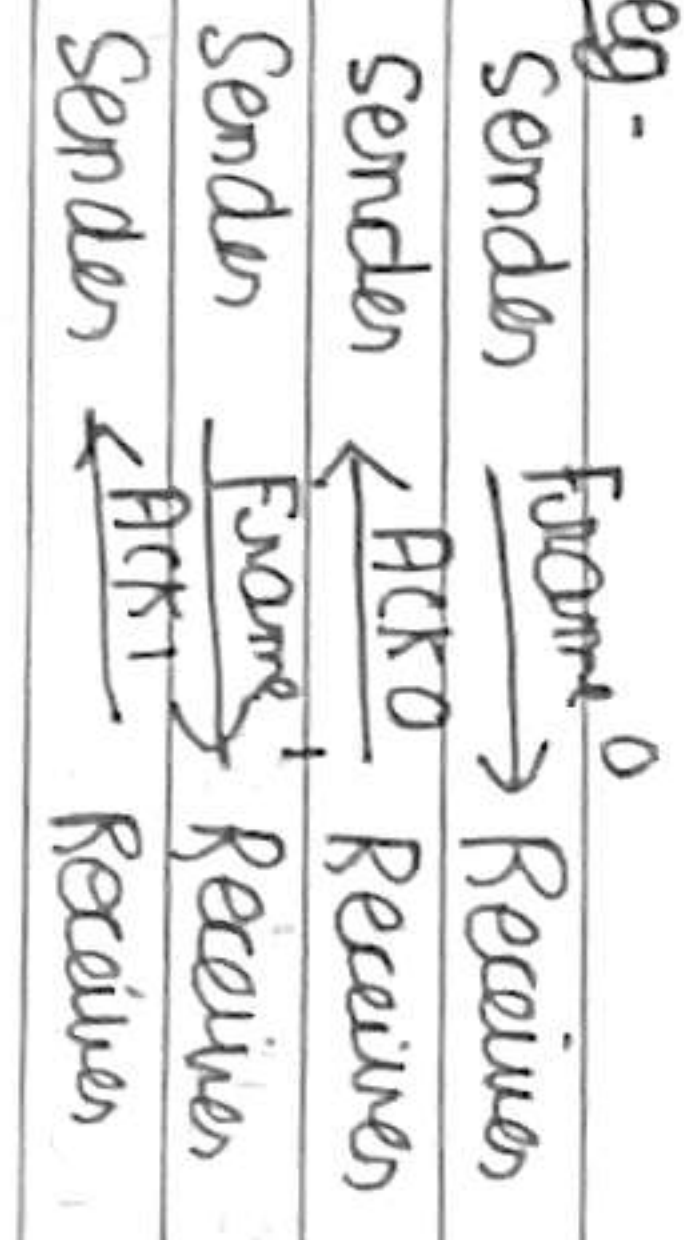
1) Stop & Wait Protocol -

- Sender transmits one frame & waits for an acknowledgement (ACK) before sending the next
- Flow control: ensures the sender does not overwhelm the receiver
- Error control: If an ACK is not received within a time out, the sender retransmits the frame

Advantages:

- Disadvantages: Inefficient for long distance or high speed links

-eg-



2) Go-Back-N ARQ (Automatic Repeat Request)

- Sender can transmit multiple frames (up to a window size N) without waiting for an ACK.
- Flow control: managed by sliding window
- Error control: If error is detected in a frame receiver discards the frame & all subsequent frames. Sender retransmits from the wrong frame onwards.
- Advantages: Higher efficiency than stop & wait
- Disadvantages: Wastes bandwidth due to transmission of multiple frames even if only one had an error
- eg- Sender: [0][1][2][3][4] ---> Receiver discards [2][3][4] Sender retransmits from [2]

3) Selective Repeat ARQ

- Sender transmits multiple frames within a window size. Receiver stores correctly received frames even if out of order
- Error control: only lost frames are retransmitted, not entire sequence
- Advantages: More efficient than Go-Back N because it avoids unnecessary retransmissions
- Disadvantages: Requires more complex receiver logic & buffering
- eg: Sender --- [0][1][2][3][4] --> Error at [2] Receiver accepts [0][1] and [3][4] Sender retransmits only [2]

4) Sliding Window Protocol

- Sender maintains a window of frames it can send before receiving ACK. Receiver also maintains a window of frames it can accept.
- Flow control: controls how many frames can be sent before ACK
- Error control: often used with ARQ mechanisms (Go-Back-N or selective repeat)
- ACKs advance the sender's window
- eg: Window size = 4
Sender can ~~receive~~ send [0,1,2,3] before ACKs.
ACK(0) received --> window slides --> Next frame (4) allowed

5) Piggybacking

- In a 2 way communication, ACKs are sent separately. Instead, they are attached to outgoing data frames to save bandwidth.
- Advantages: More efficient
- Disadvantages: Delay in pending ACK if the receiver has no data to send back
- eg: Station A --- [Data + ACK for B] ---> Station B
Station B --- [Data + ACK for A] ---> Station A

Summary table-

	Protocol	Flow control	Error control	Efficiency	Complexity
1)	Stop and wait	Yes	Yes (time out)	Low	Simple
2)	Go Back-N ARQ	Yes	Retransmits from error	Moderate	Medium
3)	Selective repeat ARQ	Yes	Retransmits only erroneous frames	High	High
4)	Sliding window	Yes	Works with ARQ	High	Medium
5)	Piggybacking	Yes	Indirectly	High	Medium

Switching

- Process of transferring data packets from one device to another device within the same network or across networks using switches
- Happens at the Data Link Layer (Layer 2) right after the Network Layer generates data packets
- Switch is a device that connects multiple devices in a network
- Ensures data from different devices does not interfere
- Works like a traffic controller by sending data packets to the correct port
- Sends data directly to connected devices
- For other networks it forwards data to a router for delivery

Network Switching

- Specific type of switching, used in computer networks
- It is about transferring data packets from one device to another across a network

Hop = 30
limit

Steps:

Network's
Statistical
Look up

- Frame reception: switch receives a frame from a device
 - MAC address extraction: reads the destination MAC address from the frame
 - Lookup: searches its switching table for the destination MAC address
 - Forwarding: If found, sends the frame to the correct port. If not found, uses flooding (sends to all ports except incoming one)
- (iv) Table update: learns new MAC addresses & updates the table for future forwarding

(vi) Frame: Forwards the frame to the correct destination device/network

Types of switching

1) Message switching -

- Entire data block/message is forwarded across the entire network thus, making it highly inefficient

- Key Features:

(a) Store and Forward:

Each intermediate node must store the entire message before forwarding it

(b) Message Delivery:

Message are transmitted as complete units

Communication is held up until message reaches its destination

(c) Network Components:

All consists of:

- Transmission links (channels)
- Store and forward switching nodes
- End stations (source & destination devices)

2) Circuit switching -

- A dedicated connection is established between sender & receiver before communication starts
- Connection receives the complete bandwidth of the network until data is transferred completely.
- (i) Bandwidth is divided into pieces
- (ii) Bit delay is constant
- (iii) Data can flow without delay once circuit is established

Methods:

- FDM
- TDM

-Phase:

- (i) Circuit establishment
- (ii) Data transfer (between source & destination)
- (iii) Circuit disconnection

3) Packet switching -

- Data divided into small packets that travel independently through the network using available paths
- Each packet carries its destination address.
- At ~~dest~~ destination, packets are reassembled into original message.

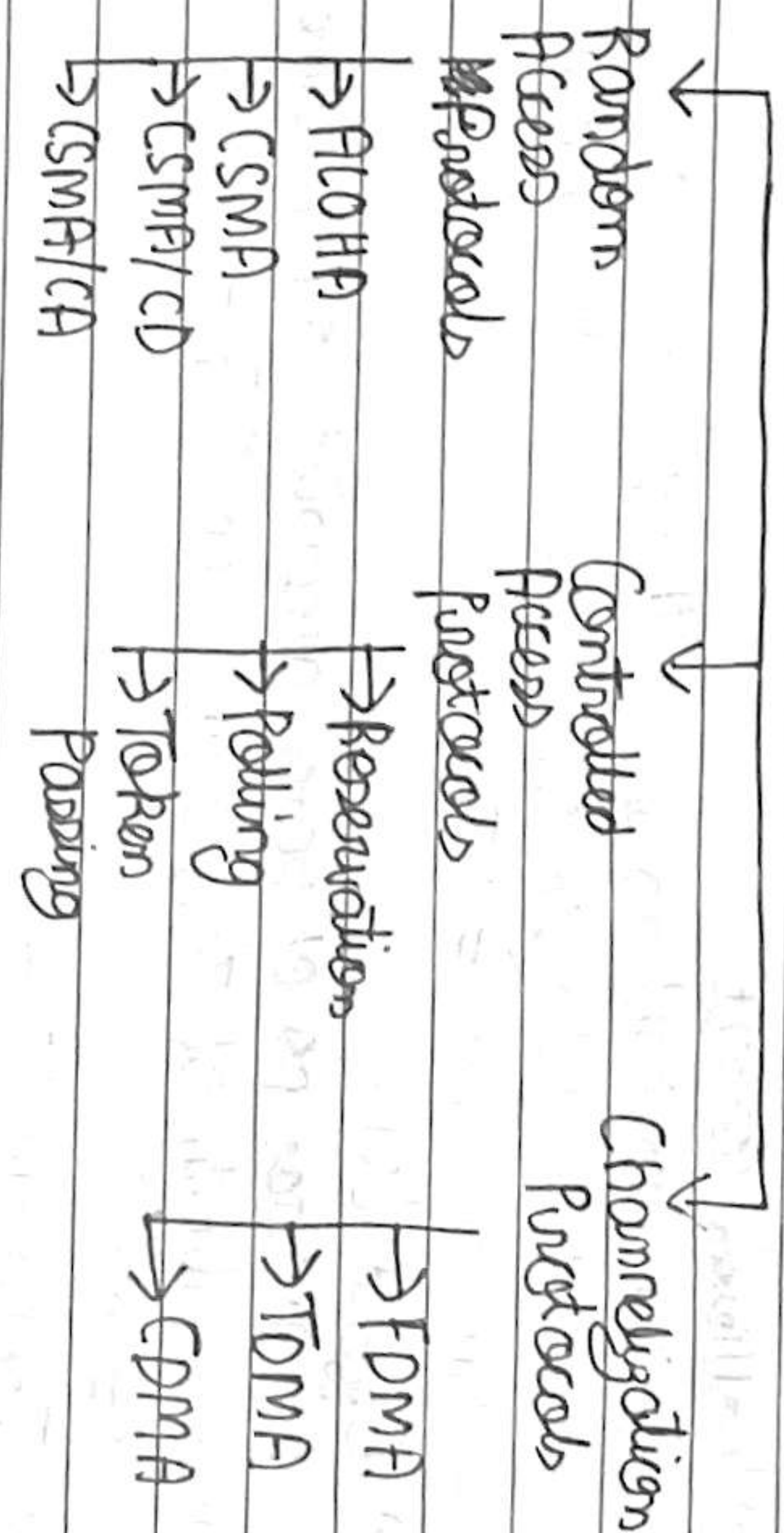
Comparison Summary -

Feature	Message	Circuit	Packet
1) Path	None	Predefined	Dynamic / Virtual
2) Efficiency	Very Low	Low	High
3) Delay	High	Low	Variable
4) Reliability	Low	High	High
5) Eg.	Telegraph	Landline cable	Internet

Multiple Access Protocols -

- Used to control how data is transmitted when multiple devices are trying to communicate over the same network.
- These protocols ensure that data packets are sent and received efficiently, without collisions or interference
- They help manage the network traffic so that all devices can share the communication channel smoothly and effectively.

Multiple Access Protocols



1) Random Access Protocols -

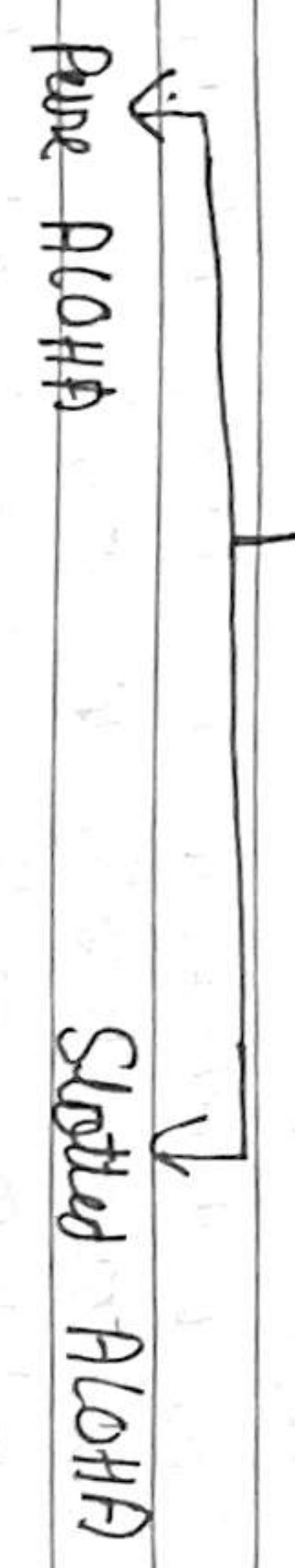
- All stations have same opportunity that is no station has more priority than another station
- Any station can send data depending on medium's state.
- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

(a) ALOHA -

- Designed for wireless LAN but also applicable for shared medium
- ~~Designed~~ Multiple stations can transmit data at the

same time & hence lead to collision & data being garbled.

ALOHA



Pure ALOHA -

- 1) Working Principle -
- A station sends data & waits for an ACK
- If No ACK is received within a certain time, it waits for a random back off time (T_b) & retransmits
- 2) Collision concept -
- A collision happens if another frame starts during a window of length $2T_f$ (before or during transmission)
- Vulnerable time = $2T_f$

3) Offered Load (G) -

G = Average no. of frames generated per frame time

4) Probability of Zero collision in $2T_f$ -

$$P = e^{-2G}$$

5) Throughput (S) -

$$S = G e^{-2G}$$

6) Maximum Throughput:

- Occurs when $G = 0.5$

$$S_{max} = 0.5 \times e^{-1} \approx 0.184$$

- Efficiency = 18.4% of channel capacity

Slotted ALOHA -

- 1) Working Principle -
- Similar to Pure ALOHA but time is divided into equal slots

- Stations can transmit only at the beginning of a slot

- If a station misses a slot, it waits for the next one - reducing collision

2) Collision concept -

- Collision occurs if another frame starts within the same slot

- Vulnerable Time = Frame Transmission Time (T_f)

3) Probability of successful transmission -

$$P_{success} = e^{-G}$$

4) Throughput (S) -

$$S = G e^{-G}$$

5) Maximum Throughput

- Occurs when $G = 1$

$$S_{max} = 1 \times e^{-1} \approx 0.368$$

- Efficiency = 36.8%

Comparison summary -

Feature	Pure ALOHA	Slotted ALOHA
1) Time Division	No time slots	Time is divided into equal slots
2) Transmission Time	Can start anytime	Can start only at the beginning of a slot
3) Vulnerable Time	$2 \times T_f$	T_f
4) Maximum Throughput	18.4%	36.8%
5) Collision	Higher	Lower
6) Reliability	Less efficient	More efficient
7) Synchronization	Not required	Required

CSMA-

1) Concept:

Stations sense the channel before transmitting data to reduce collisions

2) Working:

- If current is idle, station sends data

- If channel is busy, it waits until it becomes idle

3) Collision Possibility:

- Collisions can still occur due to propagation delay

4) Example:

- Station starts transmitting after sensing idle channel

- Before A's signal reaches other, station B also senses idle & transmits $\rightarrow$ collision occurs

5) Vulnerable Time:

- Equal to propagation time

Controlled Access Protocols -

CSMA Access Methods



(i) 1-Persistent:

- Continuously senses the channel
- Sends immediately when channel becomes idle
- Can cause more collisions

(ii) Non Persistent:

- Waits a random time before rechecking if busy
- Reduces collisions but increases delay

(iii) P-Persistent:

- Transmits with probability p when channel idle
- Waits & retries with same rule until success
- Used in Wi-Fi & packet radios

(iv) 0-Persistent:

- Transmission order pre decided
- Node transmits only in its assigned slot

CSMA/CD -

- Detects collisions during transmission
- Transmission stops immediately when collision is detected
- Used mainly in wired networks

CSMA/CA -

- Avoid collisions before they occur
- Sender checks channel & uses acknowledgements
- Used in wireless networks (Wi-Fi) where detecting collisions is difficult
- Interframe Space (IFS) -
 - Waits briefly after channel is idle before sending
 - Prevents collisions from propagation delay
- Contention Window -
 - Time divided into slots; sender selecting picks random
 - Wait time doubles if channel remains busy slot to wait
 - Times resumes once channel is idle again
- Acknowledgement: Sender transmits if ACK not received before timeout

Logical Address/Internet Protocol-#

IPv4-

- Most widely used system for identifying devices on a network
- Uses a set of 4 numbers, separated by periods (like 192.168.0.1) to give each device a unique address.

- Range: 1-255

- Serves 2 main purposes:

- (i) Identification: uniquely identifies a device on a network
- (ii) Location addressing: indicates where a device is located within a network, making data routing possible

- Parts of IPv4:

(i) Network Part:

Indicates the distinctive variety that's appointed to the network. Identifies the category of the network that's changed

(ii) Host Part:

Uniquely identifies the machine on your network.

(iii) Subnet Mask:

Non obligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets.

- Types of classes:

- Class A: 1-126
- Class B: 128-191
- Class C: 192-223
- Class D: 224-239
- Class E: 240-255

127 = used to check the local host connection (loopback address)

- Class A
- Class B
- Class C
- Class D
- Class E

IPv6-

- With 128 bit address space, it allows 340 undecillion unique address space

- Consists of 8 groups of 4 hexadecimal digits separated by ':' and each hex digit representing 4 bits so total length of IPv6 is 128 bits

- First 48 bits represent Global Routing Prefix

- Next 16 bits represent Student ID

- Last 64 bits represent the host ID

- First 64 bits represent the network portion & the last 64 bits represent the interface id.

- Global Routing Prefix:

Portion of the address used within an IPv6 address that is used to identify a specific network or subnet within the larger IPv6 internet.

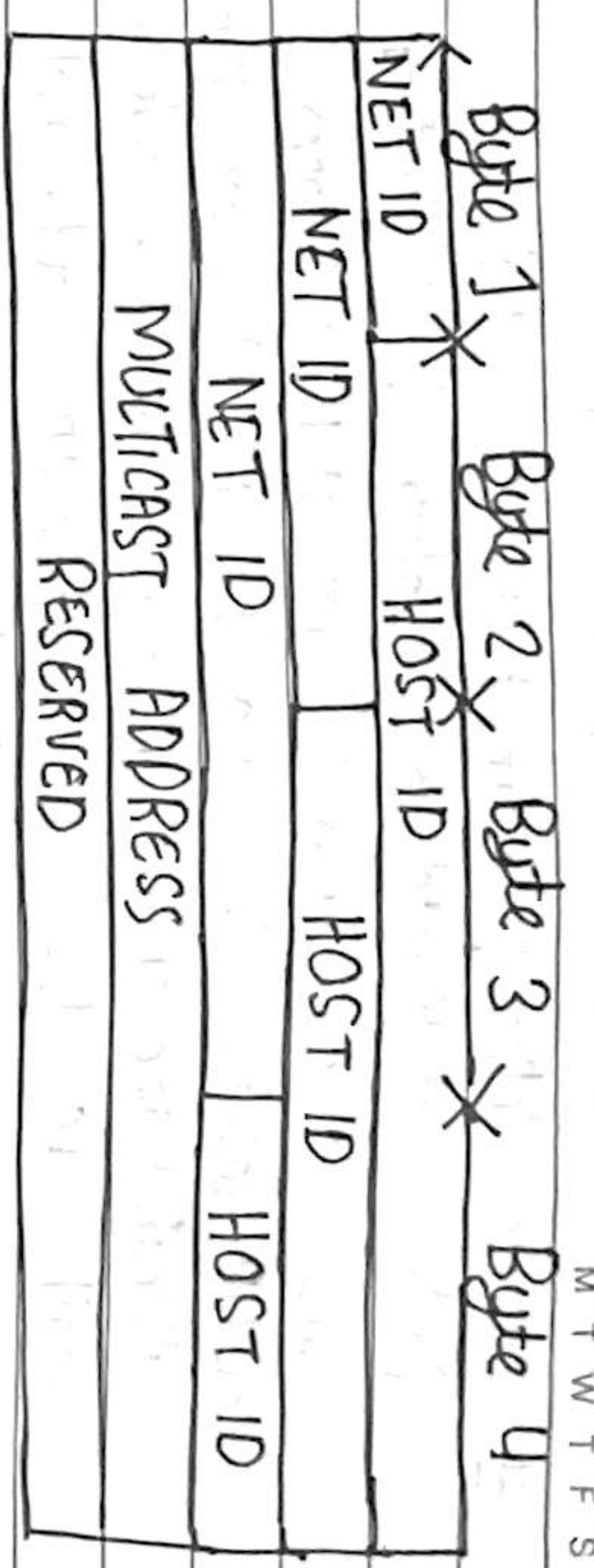
Assigned by an ISP or regional internet registry (RIR)

- Student ID:

Portion of the address used within an organization to identify subnets. Usually follows Global Routing Prefix.

- Host ID:

Last part of the address, is used to identify a specific host on a network



IP addresses = Plets of land
 All system: Fixed (Class A= huge, B=medium, C=small)
 CIDR: cut land in the size you exactly

Classless Inter Domain Routing (CIDR) -

- Method of IP address allocation & IP routing that allows for more efficient use of IP addresses.
- CIDR based on the idea that IP addresses can be allocated and routed based on their network prefix rather than their class, unlike the traditional way of IP address allocation.

- Subnet Masks:

- Class A: NID, HID, HID, HID = (255.0.0.0) = 8 bits
- Class B: NID, NID, HID, HID = (255.255.0.0) = 16 bits
- Class C: NID, NID, NID, HID = (255.255.255.0) = 24 bits
- CIDR value
- Class A: 8 bits
- Class B: 16 bits
- Class C: 24 bits

Address Mapping -

- Essential for translating between different types of addresses in a network, such as IP addresses & MAC addresses.
- Key protocols:
- (i) ARP (Address Resolution Protocol)
- (ii) RARP (Reverse Address Resolution Protocol)
- (iii) BOOTP (Bootstrap Protocol)
- (iv) DHCP (Dynamic Host Configuration Protocol)

#	Comparison:	Addressing	Mapping	Mapping	Protocol
Aspect	ARP	RARP	BOOTP	DHCP	
1) Main Function	Maps IP address to MAC address	Finds IP address from MAC address	Assigns static IP and config	Assigns dynamic IP and full network config	
2) Address mapping	IP → MAC	MAC → IP	Server → Client	Server → Client	
3) Used for	Finding physical address in LAN	Receiving device without IP	Providing IP during boot	Automating IP and config management	
4) Config type	Most applicable	Static	Static	Dynamic	
5) Info provided	MAC address	IP address	IP gateway, subnet mask	IP, gateway, subnet mask, DNS	
6) Comm. Scope	LAN	LAN	Across networks	Across networks	
7) Protocol Type	Resolution	Resolution	Configuration	Configuration	

Forwarding & Unicast Routing Protocols

1) Routing -

- When a device has multiple paths to reach a destination it selects one preferred path - this process is called Routing
- Routers (layer 3 devices in OSI model) perform routing
- A default route is configured to specify where to send packets when no specific route is found in the routing table

2) Unicast Routing Protocols -

- (a) Distance Vector Routing Protocol
- Makes routing decisions based on no. of hops between source & destination

2) Unicast Routing -

- Unicast traffic is data sent from one source to a specific destination
- Most internet traffic is unicast
- Router looks up the routing table & forwards the packet to the next hop
- It's the simplest form of routing since destination is known

3) Unicast Routing Protocols -

(a) Distance Vector Routing Protocol:

- Makes routing decisions based on no. of hops between source & destination
- Route with fewer hops = best route
- Each router advertises its best routes to neighbors
- Over ~~and~~ time, all routers learn the entire network topology

- Eg. Routing Information Protocol (RIP)

(b) Link State Routing Protocol:

- Considers the state of network links of all routers
- Each router builds a complete network map (graph)
- Then, every router independently calculates the shortest path to each destination.
- Eg. Open Shortest Path First (OSPF)
- Intermediate System to Intermediate System (IS-IS)

4) Broadcast Routing -

- By default, routers do not forward broadcast packets - they separate broadcast domains
- However, broadcast forwarding can be enabled in special cases
- Broadcast messages are sent to all devices on the network

5) Multicast Routing -

- A special form of broadcast routing
- In broadcast routing, packets go to all nodes (even unwanted ones)
- In multicast routing, packets go only to interested nodes
- Eg. Distance Vector Multicast Routing Protocol (DVMRP)
- Multicast Open Shortest Path First (MOSPF)

6) Classification of Routing Algorithms -

- Routing algorithms are broadly divided into:
 - (i) Flooding
 - (ii) Shortest Path Algorithms
 - Dijkstra's Algorithm
 - Bellman-Ford Algorithm
 - Floyd-Warshall Algorithm

Summary -

- 1) Unicast $\rightarrow$ One to one communication
- 2) ~~Multiple~~ Broadcast $\rightarrow$ One to all communication
- 3) Multicast $\rightarrow$ One to ~~many~~ many (interested nodes only)
- 4) Routing protocols decide the best path using hop count (Distance Vector) or link states (Link State)